

Unsupervised Range-Nullspace Learning Prior for Multispectral Images Reconstruction

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Abstract—Snapshot Spectral Imaging (SSI) techniques, with the ability to capture both spectral and spatial information in a single exposure, have been found useful in a wide range of applications. SSI systems generally operate within the ‘encoding-decoding’ framework, leveraging the synergism of optical hardware and reconstruction algorithms. Typically, reconstructing desired spectral images from SSI measurements is an ill-posed and challenging problem. Existing studies utilize either model-based or deep learning-based methods, but both have their drawbacks. Model-based algorithms suffer from high computational costs, while supervised learning-based methods rely on large paired training data. In this paper, we propose a novel Unsupervised range-Nullspace learning (UnNull) prior for spectral image reconstruction. UnNull explicitly models the data via subspace decomposition, offering enhanced interpretability and generalization ability. Specifically, UnNull considers that the spectral images can be decomposed into the range and null subspaces. The features projected onto the range subspace are mainly low-frequency information, while features in the nullspace represent high-frequency information. Comprehensive multispectral demosaicing and reconstruction experiments demonstrate the superior performance of our proposed algorithm.

Index Terms—Demosaicing, Spectral imaging, Inverse problem, Optimization, Unsupervised learning, Image reconstruction

I. INTRODUCTION

Spectral Imaging (SI) techniques have attracted significant attention owing to their ability to capture information across a wide range of wavelengths within the spectrum. SI holds great potential in various domains, including remote sensing [1][2], medical diagnosis [3], and face detection [4]. To obtain a 3-dimensional (3D) spectral image datacube, prevalent SI systems rely on scanning mechanisms such as whiskbroom or pushbroom scanners [5]. While these scanning-based imaging systems provide fine spectral resolution, they are constrained by limitations related to imaging speed, sophisticated opto-mechanical system design, and high costs.

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Recently, numerous Snapshot Spectral Imaging (SSI) methods have been proposed to enhance the efficiency of spectral imaging [6]–[10]. The fundamental principle underlying SSI is the ‘encoding-decoding’ framework. The encoding process maps a 3D spectral image datacube onto 2D or 1D space. Subsequently, decoding algorithms are employed to reconstruct the desired spectral image from the undersampled measurements. For instance, the Multi-Spectral Filter Array (MSFA)-based SSI system utilizes a pixel-wise filter array to modulate spectral image into a 2D raw image [11]. Similar to the Bayer filter used in Red-Green-Blue (RGB) cameras, each pixel (photo-receptor) in MSFA captures the response of a pre-determined band image. As a result, spectral image demosaicing and reconstruction algorithms are crucial for estimating missing pixels when the raw image is unfolded back into the complete spectral image datacube.

While algorithms for reconstructing RGB images have been extensively studied over the decades [12], spectral image restoration remains a relatively challenging research area. The assumption of high spatial and channel correlation inherent in RGB images cannot be seamlessly applied to multispectral imaging scenarios [13][14][15]. Therefore, it is imperative to develop algorithms tailored to the distinctive characteristics of spectral images and their encoding processes. One such solution is the Weighted Bilinear (WB) interpolation, which utilizes a specific convolution kernel to calculate missing pixels for each spectral band individually [15]. Additionally, Brauers *et al.* [16] introduced a Spectral Difference (SD)-based demosaicing method that considers both spatial and spectral correlations. While these traditional statistics-based methods are computationally efficient, they often fail to preserve fine details in reconstructed images.

Furthermore, Tsagkatakis *et al.* [17] introduced the concept of compressive sensing [18] to multispectral demosaicing. They formulated spectral image reconstruction as a Graph and Rank regularized Matrix Recovery (GRMR) optimization problem. Other alternative regularizations involving total variation (spatial smoothness) [19][20], low-rank prior [21][22], and untrained neural network prior [23]–[27], can be further explored for spectral image reconstruction. These algorithms, by leveraging prior knowledge, have exhibited improved performance compared to conventional methods. However, two main issues persist with iterative optimization-based spectral image demosaicing and restoration algorithms.

- *Lacking consideration of high-frequency features.* Commonly used priors do not adequately exploit high-frequency features. These priors either assume the data is smooth in the spatial domain (total variation [19]) or

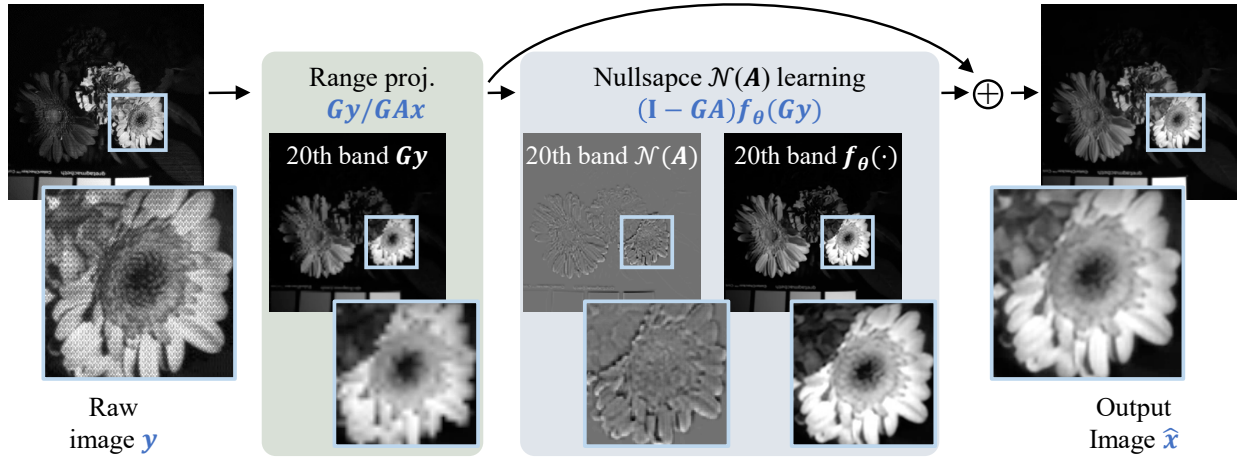


Fig. 1: The flowchart of our proposed Unsupervised range-Nullspace learning (UnNull) method for multispectral demosaicing and reconstruction. The desired spectral images are finally obtained via the summation of the data projected onto the range (Gy or GAx) and null subspaces ($\mathcal{N}(A)$ or $(I - GA)f_\theta(Gy)$), where G denotes the generalized inverse matrix of A (i.e., $AGA = A$) and $f_\theta(\cdot)$ represents learnable model used to reconstruct the spectral image for recovering high-frequency details that projected onto the nullspace.

sparse in transform domains (low-rank [21][28]). Consequently, reconstructed images might lack fine details.

- *High computational cost.* Convex programming-based methods are computationally expensive. For example, GRMR [17] requires nearly 80 seconds to reconstruct the spectral images of a single captured scene. This slow algorithm running time significantly restricts their practical applications.

More recently, researchers have turned to deep learning, particularly supervised learning methods, as an alternative to convex programming-based algorithms [29][30][31][32]. For example, Feng *et al.* [32] introduced the Mosaic Convolution-Attention Network (MCAN) for multispectral demosaicing. Furthermore, various spectral image reconstruction network architectures were presented in the 2022 New Trends in Image Restoration and Enhancement (NTIRE) challenge [33][34]. These deep learning-based approaches have exhibited superior performance and fast inference speed. However, they face challenges such as poor interpretability, dependence on large-scale paired datasets, and the need to re-train networks when optical hardware changes. As a result, developing an effective, efficient, and generalized spectral image reconstruction method remains a key challenge.

In this paper, we propose a novel Unsupervised range-Nullspace learning (UnNull) prior for multispectral demosaicing and reconstruction. Specifically, we recognize that the spectral image data can be decomposed into range and null subspaces, as illustrated in Fig. 1. The information projected onto the range space primarily consists of low-frequency features (namely global structures), while the data projected onto the nullspace typically represents high-frequency details, e.g., edges and fine textures. Low-frequency features are often preserved during compressive spectral imaging processes, while high-frequency details tend to be distorted or lost after sub-sampling. Although the concept of range-nullspace decomposition has been previously employed in natural image

reconstruction or generation [35]–[37], existing methods rely on supervised learning to solve for the range and/or nullspace. These methods often lack generalization ability and robustness. Furthermore, the projection matrix (range subspace) of SSI systems differs from the natural image scenarios, which were previously unexplored.

In summary, the main contributions are as follows.

- *A novel prior:* We introduce the UnNull prior, which explicitly decomposes spectral images into range and null subspaces. Moreover, we present a detailed theoretical analysis of UnNull for spectral image reconstruction. Leveraging this prior improves the interpretability of deep learning-based methods.
- *Unsupervised optimization strategy:* Apart from previous supervised methods, the UnNull prior is optimized in an unsupervised learning manner. By embedding interpretable prior and the imaging sensing model, our method shows superior generalization ability across unseen datasets and diverse imaging systems.
- *Improved performance and efficiency:* The proposed UnNull algorithm demonstrates leading performance in spectral image reconstruction. Furthermore, when compared to existing optimization-based methods, it offers enhanced computational efficiency.

Finally, we conduct comprehensive experiments on snapshot spectral imaging tasks, such as spectral image demosaicing and reconstruction. Results validate the superior performance of the proposed UnNull framework across different datasets, noise levels, and imaging systems.

II. RELATED WORKS

A. Model-driven Spectral Image Reconstruction Algorithms

Spectral image reconstruction algorithms aim to obtain spectral images from incomplete noisy measurements, which can be broadly categorized into two types: model-driven [38]–[42] and data-driven [29][31][43][44] methods.

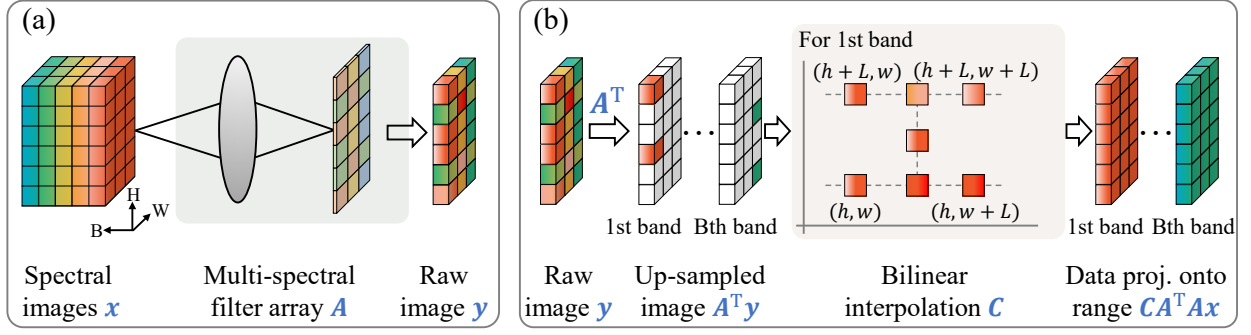


Fig. 2: (a) The forward observation model of MSFA-based spectral imaging system; (b) the formulation of the generalized inverse matrix via bilinear interpolation.

Model-driven approaches focus on leveraging prior knowledge about data and employing convex programming to reconstruct desired spectral images. For instance, the total variation (TV) prior (sparsity in the gradient domain) of data is utilized in [20], where the Generalized Alternating Projection (GAP) optimization algorithm is applied to solve the TV-regularized problem and obtain spectral images. Liu *et al.* [21] considered that the spectral image has non-local self-similarity and low-rank characteristics. The Alternating Direction Method of Multipliers (ADMM) [45] optimization framework is then leveraged to find the optimal spectral images. Tsagakaki *et al.* introduced the concept of compressive sensing into multispectral demosaicing, considering the low-rank prior and data graph structure of spectral image [17]. In addition, with the introduction of the deep image prior or untrained neural network prior [23][46], Meng *et al.* [47] proposed a deep image prior regularized optimization problem for spectral images reconstruction. Chen *et al.* [25] developed a Deep Spectral image Prior (DSP) for spectral image demosaicing. Furthermore, a combined low-rank and deep image prior was introduced in [27] for snapshot compressive imaging.

B. Data-driven Spectral Image Reconstruction Algorithms

On the other hand, supervised deep learning networks (data-driven methods) focus on learning an end-to-end function for mapping measurements to desired spectral images. Dong *et al.* [48] proposed a model-guided deep unfolding convolutional network for reconstructing hyperspectral images. Fu *et al.* [49] introduced a spectral-spectral convolutional neural network for joint spectral response function selection and spectral image recovery. Miao *et al.* [29] proposed λ -net for reconstructing HSIs from measurements. The spatial-spectral self-attention network (TSA-Net) [50] was designed for reconstructing spectral images from SSI measurements. Huang *et al.* [31] designed a learnable Gaussian scale mixture model for spectral image reconstruction. A bi-directional recurrent neural network with residual blocks and self-attention is introduced in [51] for snapshot compressive imaging. Feng *et al.* proposed an attention module-based demosaicing network [32]. Moreover, they introduced an unsupervised learning framework-based multispectral demosaicing method for further improving the model's generalization [52]. In addition, a pseudo-panchromatic image (PPI) edge-infused spatial-spectral adap-

tive residual network was proposed in [53] to enhance the extraction of high-frequency information. Recently, with the advancement of Transformers architecture, the Transformer-based spectral image reconstruction networks proposed in [54]–[57] obtained state-of-the-art results.

C. Snapshot Spectral Imaging Model

This paper adopts unified notations to enhance clarity and facilitate understanding. High-order tensors and vector spaces are represented by Euler script, *e.g.*, $\mathcal{X}$; matrices are denoted by bold uppercase letters, like $\mathbf{X}$; vectors are represented by bold lowercase letters, *e.g.*, $\mathbf{x}$; scalar data are expressed using standard uppercase or lowercase letters, such as H or h .

Let $\mathcal{X} \in \mathbb{R}^{H \times W \times B}$ denote the desired spectral image, where H and W are the spatial height and width, and B signifies the number of spectral channels. The tensor $\mathcal{X}$ (or its mode-3 representation $\mathbf{X} \in \mathbb{R}^{HW \times B}$) can be vectorized as $\mathbf{x} \in \mathbb{R}^{N_x}$, where $N_x = HWB$ is the total number of elements. The forward observation model for various Snapshot Spectral Imaging (SSI) systems can be generally unified using a linear transform model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{e}, \quad (1)$$

where $\mathbf{y} \in \mathbb{R}^{N_y}$ indicates the measured signal and its dimension is always smaller than that of the spectral image ($N_y \ll N_x$); $\mathbf{e} \in \mathbb{R}^{N_y}$ denotes the noise introduced by the imaging system; and $\mathbf{A} \in \mathbb{R}^{N_y \times N_x}$ signifies the measurement matrix of the SSI system. In this paper, we focus on the Multi-Spectral Filter Array (MSFA)-based spectral imaging system [11], as shown in Fig. 2(a). However, the proposed methods can be easily extended to other SSI techniques.

For the MSFA spectral imaging system, the sensing matrix $\mathbf{A} \in \mathbb{R}^{HW \times HWB}$ can be formulated as

$$\mathbf{A} = [\text{diag}(\text{vec}(\mathbf{M}_1)), \dots, \text{diag}(\text{vec}(\mathbf{M}_B))], \quad (2)$$

where $\text{diag}(\cdot)$ denotes the diagonal matrix of a vector, $\text{vec}(\cdot)$ represents the vectorization operation, and $\mathbf{M}_B \in \mathbb{R}^{H \times W}$ is the sampling matrix for the B -th band, given by

$$\mathbf{M}_B(h, w) = \begin{cases} 1, & \text{if } (h, w) \in \Omega_B \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where Ω_B represents the set of measured pixels for the B -th channel, meaning that each pixel only captures a dominant

spectral band. In this setup, the number of spectral bands B is determined by the filter array size $L \times L$, where $B = N_F = L^2$. For example, with a 4×4 filter array, $B = 16$.

Beyond the task of multispectral demosaicing, we address a more challenging problem—spectral image reconstruction. Specifically, by incorporating the spectral response function of filters, the matrix in Eq. (2) is defined as

$$\mathbf{M}_B(h, w) = t_B^{hw}, \quad (4)$$

where t_B^{hw} signifies the B -th spectral transmittance of the filter corresponding at the pixel (h, w) . In this case, the number of spectral bands B can exceed the number of filters ($B \geq N_F = L^2$). For instance, a spectral image with 31 bands can be reconstructed using a multispectral camera equipped with a 5×5 filter array.

Although the sensing matrices differ between the multispectral demosaicing and multispectral reconstruction tasks, we demonstrate that the proposed method is applicable to both cases. This is because the sensing matrix for multispectral reconstruction task can be decomposed into a binary sensing matrix $\mathbf{A}_B \in \mathbb{R}^{HW \times HW N_F}$ (identical to that used in multispectral demosaicing) and a filter response matrix $\mathbf{F} \in \mathbb{R}^{HW N_F \times HW B}$, such that

$$\mathbf{A} = \mathbf{A}_B \times \mathbf{F}. \quad (5)$$

In conclusion, the forward observation model is rewritten as

$$\mathbf{Y} = \sum_{b=1}^B \mathcal{X}(:, :, b) \odot \mathbf{M}_b + \mathbf{E}, \quad (6)$$

where $\mathbf{Y} \in \mathbb{R}^{H \times W}$ is the measured raw image and $\mathbf{E}$ denotes the noise of imaging systems. Multispectral demosaicing and reconstruction methods aim to obtain $\mathcal{X}/\mathbf{X}$ given $\mathbf{Y}$ and $\mathbf{A}$.

III. PROPOSED METHOD

Motivation. Recent studies have highlighted the potential of range-nullspace ($\mathcal{R}$ - $\mathcal{N}$) decomposition in image reconstruction and generation [35][36][37]. For instance, Davies *et al.* [35] employed two neural networks to separately approximate the range and null subspaces. Wang *et al.* [36] leveraged a pre-trained Generative Adversarial Network (GAN) for generating components in the nullspace. Building on the concept of $\mathcal{R}$ - $\mathcal{N}$ decomposition, Chen *et al.* [37] designed a range-nullspace decomposition-based unrolling network for inverse imaging, which considers the physical information in training and deploying neural networks.

However, these approaches are built on supervised learning frameworks and require large-scale datasets, limiting their ability to generalize across diverse SSI systems or unseen datasets. Moreover, the application of $\mathcal{R}$ - $\mathcal{N}$ decomposition in spectral images reconstruction remains largely unexplored, especially given the increased complexity of the forward observation model in spectral imaging (Eq. (1)–Eq. (6)).

A. Unsupervised Range-Nullspace Learning for Spectral Images Reconstruction

In this paper, we propose a novel unsupervised $\mathcal{R}$ - $\mathcal{N}$ decomposition-based method for spectral image reconstruction.

The desired spectral images are obtained by summing components derived from both the range and nullspace. To establish the foundation, we first introduce key concepts of $\mathcal{R}$ - $\mathcal{N}$ decomposition.

Definition 1 (Generalized Inverse Matrix). *Given a linear transformation matrix $\mathbf{A} \in \mathbb{R}^{N_y \times N_x}$, a matrix $\mathbf{G} \in \mathbb{R}^{N_x \times N_y}$ is called a generalized inverse of $\mathbf{A}$ if it satisfies*

$$\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}. \quad (7)$$

Remark 1. *For any matrix $\mathbf{A}$, its generalized inverse matrix $\mathbf{G}$ always exists, although it might not be unique. Furthermore, $\mathbf{G}$ can vary depending on the structure of $\mathbf{A}$.*

Definition 2 (Range $\mathcal{R}$). *Let the generalized inverse matrix $\mathbf{G} : \mathcal{Y} \rightarrow \mathcal{X}$ be a linear transformation. The range of $\mathbf{G}$ is defined as*

$$\mathcal{R}(\mathbf{G}) = \{\mathbf{G}\mathbf{y} \mid \forall \mathbf{y}\}. \quad (8)$$

In other words, the range of $\mathbf{G}$ is the set of all outputs produced by $\mathbf{G}$ for any input $\mathbf{y}$, which corresponds to the column space of $\mathbf{G}$.

Definition 3 (Nullspace $\mathcal{N}$). *Let $\mathbf{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be a linear transformation. The nullspace of $\mathbf{A}$ is defined as*

$$\mathcal{N}(\mathbf{A}) = \{\mathbf{x} \mid \mathbf{A}\mathbf{x} = \mathbf{0}\}. \quad (9)$$

That is, the nullspace consists of all vectors $\mathbf{x}$ in $\mathcal{X}$ that $\mathbf{A}$ maps to zero vector.

Definition 4 ($\mathcal{R}$ - $\mathcal{N}$ Decomposition). *Given $\mathcal{R}(\mathbf{G}) \oplus \mathcal{N}(\mathbf{A}) = \mathbb{R}^{N_x}$ be the vector space $\mathbb{R}^{N_x}$ where $\oplus$ denotes the direct sum, for any data sample $\mathbf{x} \in \mathbb{R}^{N_x}$, the range-nullspace decomposition can be formulated as*

$$\mathbf{x} = \text{proj}_{\mathcal{R}} \mathbf{x} + \text{proj}_{\mathcal{N}} \mathbf{x}, \quad (10)$$

where $\text{proj}_{\mathcal{R}} \cdot$ and $\text{proj}_{\mathcal{N}} \cdot$ represents the projections of $\mathbf{x}$ onto the range $\mathcal{R}(\mathbf{G})$ and nullspace $\mathcal{N}(\mathbf{A})$, respectively.

Based on the observation that the idea sensing matrix of the MSFA-based spectral imaging system can be interpreted as a skip downsampling operation applied to each band, here, we construct the generalized inverse matrix $\mathbf{G}$ using bilinear interpolation, as depicted in Fig. 2(b).

Lemma 1. *The bilinear interpolation serves as a generalized inverse of the sensing matrix $\mathbf{A}$ in the MSFA-based spectral imaging system.*

Proof. We begin by factorizing the generalized inverse matrix $\mathbf{G}$ into two components,

$$\mathbf{G} = \mathbf{C}\mathbf{A}^\top, \quad (11)$$

where $\mathbf{A}^\top \in \mathbb{R}^{N_x \times N_y}$ signifies the transpose of matrix $\mathbf{A}$, and $\mathbf{C} \in \mathbb{R}^{N_x \times N_x}$ represents the matrix form of bilinear interpolation. In particular, the bilinear interpolation is implemented as a convolution operation, where the convolution kernel $\mathbf{W} \in \mathbb{R}^{(2L-1) \times (2L-1)}$ is defined as

$$\mathbf{W} = \mathbf{h}^\top \mathbf{h}, \quad (12)$$

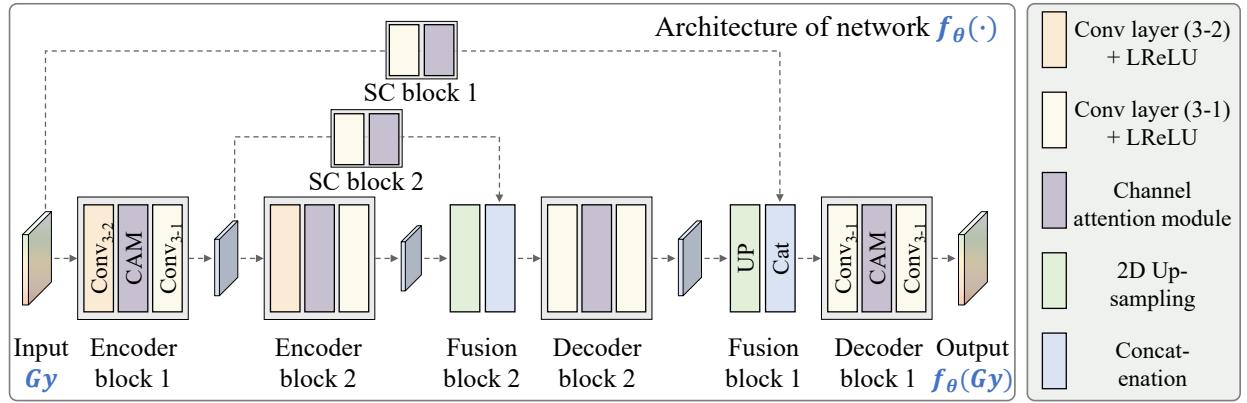


Fig. 3: The designed lightweight network $f_{\theta}(\cdot)$ for nullspace learning, which follows the U-shape network architecture.

and $\mathbf{h} \in \mathbb{R}^{1 \times (2L-1)}$ is given by

$$h = \text{conv}(\mathbf{1}_L, \mathbf{1}_L)/L, \quad (13)$$

where $\mathbf{1}_L \in \mathbb{R}^{1 \times L}$ is an all-ones vector and $\text{conv}(\cdot, \cdot)$ represents the convolution operation between two vectors. The matrix C is then expressed as a circulant matrix

$$C = \begin{bmatrix} W_{00} & \dots & W_{0(2L-1)} & 0 & \dots & 0 & W_{10} & \dots & W_{1(2L-1)} & 0 & \dots & 0 & \dots \\ 0 & W_{00} & \dots & W_{0(2L-1)} & 0 & \dots & 0 & W_{10} & \dots & W_{1(2L-1)} & 0 & \dots & \dots \\ 0 & 0 & W_{00} & \dots & W_{0(2L-1)} & 0 & \dots & 0 & W_{10} & \dots & W_{1(2L-1)} & \dots & \dots \\ & & & & & & \dots & & & & & & \\ 0 & \dots & 0 & \dots & W_{00} & \dots & W_{0(2L-1)} & 0 & \dots & 0 & W_{10} & \dots & W_{1(2L-1)} \end{bmatrix}. \quad (14)$$

To ensure the dimensions remain consistent, the vector to be multiplied by C is padded with zeros. Since the bilinear interpolation operator preserves the original data, the following holds

$$AGAx = AGy = y = Ax, \quad (15)$$

which implies $\mathbf{AGA} = \mathbf{A}$. Thus, the bilinear interpolation is the generalized inverse matrix of the idea sensing matrix. $\square$

Lemma 2. *The matrix $\mathbf{GA} \in \mathbb{R}^{N_x \times N_x}$ is a projection matrix.*

Proof. A matrix $\mathbf{P}$ is called a projection matrix if it satisfies $\mathbf{P} = \mathbf{P}^2$. From the relationship $\mathbf{AGA} = \mathbf{A}$, we deduce

$$GA = G(AGA) = (GA)(GA) = GA. \quad (16)$$

This implies $\mathbf{GA} = (\mathbf{GA})^2$, proving that $\mathbf{GA}$ is a projection matrix. $\square$

Lemma 3. *The range of $\mathbf{GA}$ equals to the range of $\mathbf{G}$, i.e., $\mathcal{R}(\mathbf{GA}) = \mathcal{R}(\mathbf{G})$.*

Proof. For any $x \in \mathcal{R}(G)$, there exists one y such that makes $x = Gy = GAx \in \mathcal{R}(GA)$. Moreover, for $x \in \mathcal{R}(GA)$, there exists one y such that $x = GAx = G(AGAx) \in \mathcal{R}(G)$. Therefore, GA projects data onto the range of G . $\square$

Lemma 4. *The matrix $(\mathbf{I}_{N_x} - \mathbf{GA})$ corresponds to the nullspace of $\mathbf{A}$, i.e., $\mathcal{N}(\mathbf{A})$.*

Proof.

$$\mathbf{A}(\mathbf{I}_{N_t} - \mathbf{G}\mathbf{A}) = \mathbf{A} - \mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}. \quad (17)$$

where $\mathbf{I}_{N_x} \in \mathbb{R}^{N_x \times N_x}$ represents the identity matrix.

Based on the above discussion, the $\mathcal{R}\text{-}\mathcal{N}$ decomposition Eq. (10) can be expressed as

$$\mathbf{x} = \mathbf{G}\mathbf{A}\mathbf{x} + (\mathbf{I}_{N_x} - \mathbf{G}\mathbf{A})\mathbf{x}. \quad (18)$$

In the ideal case, *i.e.*, the system noise is zero ($e = 0$), the above equation can be rewritten as

$$\mathbf{x} = \mathbf{G}\mathbf{y} + (\mathbf{I}_{N_x} - \mathbf{G}\mathbf{A})\mathbf{x}. \quad (19)$$

The first term of the right-hand side ($G\mathbf{y}$) can be calculated analytically. However, since $\mathbf{x}$ is unknown during the reconstruction process, in this paper, we propose using an untrained neural network to estimate the nullspace component. Specifically, we approximate $\mathbf{x}$ with $\mathbf{z}$ such that

$$\begin{aligned}\hat{\mathbf{x}} &= \mathbf{G}\mathbf{y} + (\mathbf{I}_{N_x} - \mathbf{G}\mathbf{A})\mathbf{z} \\ &= \mathbf{G}\mathbf{y} + (\mathbf{I}_{N_x} - \mathbf{G}\mathbf{A})\mathbf{f}_\theta(\mathbf{G}\mathbf{y}),\end{aligned}\quad (20)$$

where $\hat{x}$ denotes the final output spectral images, $z = f_{\theta}(Gy)$ is the output of the neural network $f_{\theta}(\cdot)$ with parameters θ .

B. Optimization of Untrained Neural Network

Different from previous works, this paper finds the nullspace in an unsupervised learning manner. According to Eq. (19), we have

$$A\hat{x} = A[Gy + (I_{N_x} - GA)f_\theta(Gy)]. \quad (21)$$

Therefore, we can define the loss function for optimization of the untrained neural network $\mathbf{f}_{\theta}(\cdot)$ via calculating the error between $\mathbf{A}\hat{\mathbf{x}}$ and the measurement $\mathbf{y}$, *i.e.*,

$$\begin{aligned}\hat{\theta} &= \arg \min_{\theta} \|\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}\|_p \\ &= \arg \min_{\theta} \|\mathbf{y} - \mathbf{A}[\mathbf{G}\mathbf{y} + (\mathbf{I}_{N_x} - \mathbf{G}\mathbf{A})\mathbf{f}_{\theta}(\mathbf{G}\mathbf{y})]\|_p,\end{aligned}\quad (22)$$

where $\|\cdot\|_p$ represents the L_p norm. Typically, the choice of the L_p norm as the data fidelity term depends on the assumed distribution of noise $p(e)$. For example, L_2 norm is suitable for Gaussian noise, while L_1 norm is generally used for sparse noise. In this paper, we empirically present that the selection of L_p norm has a significant impact on results, especially when the measurement contains noise. Details can be found in Sec. IV C.

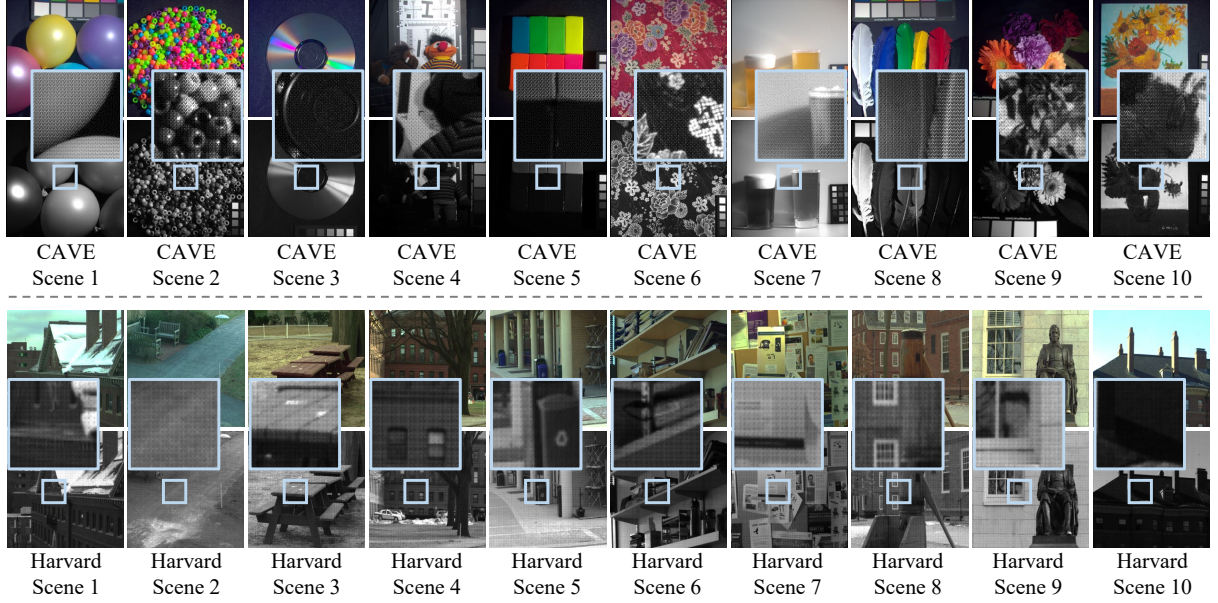


Fig. 4: Reference RGB images and the corresponding simulated 5×5 SSI raw images of testing scenes from the CAVE dataset (top) and the Harvard dataset (bottom).

Moreover, the solution of Eq. (21) is trivial and we add the spectral total variation (STV) term to regularize the nullspace learning

$$\hat{\theta} = \arg \min_{\theta} \|\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}\|_p + \lambda_R \text{STV}(\hat{\mathbf{x}}), \quad (23)$$

where λ_R is a balancing factor. From Eq. (22), it is evident that we do not need ground truth $\mathbf{x}$ for training and learning the nullspace. In the end, a stochastic gradient descent algorithm, specifically the ADAM optimizer [58], is utilized in this paper to train the network parameters.

In a nutshell, we introduce the range-nullspace learning for spectral images reconstruction in Sec. III.A. The optimization of nullspace is presented in Sec. III.B. It is obvious that the proposed Unsupervised range-Nullspace learning (UnNull) method is adaptive for each captured scene. Following the philosophy of deep image prior [23], the network $\mathbf{f}_{\theta}(\cdot)$ should be optimized for each measurement. Therefore, we design a lightweight network architecture, considering the balance between reconstruction performance and computational cost.

C. Lightweight Network Architecture of UnNull

Here, we provide details about the architecture of the neural network $\mathbf{f}_{\theta}(\cdot)$. The design principles include two key aspects. Firstly, ensuring that the input ($\mathbf{G}\mathbf{y} \in \mathbb{R}^{N_x}$) and the output ($\mathbf{z} \in \mathbb{R}^{N_x}$) have the same size. The second one is that the network should be lightweight to facilitate an efficient training process. With this in mind, we design a network architecture as depicted in Fig. 3. The network follows a U-shape framework (*i.e.*, encoder network-decoder network), which ensures that the input and output have the same data size. Specifically, the encoder includes two blocks. Each block consists of two convolutional layers followed by a Leaky Rectified Linear Unit (LReLU) activation function, and a Channel Attention Module (CAM) [59]. The first convolutional layer with a kernel size of

3 and stride of 2 (denoted as 3-2 in Fig. 3) is used for capturing spatial features and the kernel size and stride of the second convolutional layer is set to 3 and 1, respectively. The CAM is leveraged to capture spectral (channel) features. Accordingly, the decoder has two blocks, each comprising two convolutional layers (with a kernel size of 3 and stride of 1) followed by a LReLU activation function, and a CAM. In addition, the Skip Connection (SC) block that consists of a convolutional layer followed by an LReLU activation function and a CAM, and the concatenation operation are utilized to fuse feature maps from different levels. The number of channels of each feature map is set to N_c (the effect of N_c is discussed in Sec. IV.B). It is obvious that the designed network is lightweight and has a compact network size.

IV. EXPERIMENTS

Datasets. Experiments on spectral image demosaicing and reconstruction are primarily conducted using the generalized assorted pixel camera (CAVE) dataset [60] and the Harvard dataset [61]. The CAVE dataset consists of indoor laboratory scenes, and its spectral images have a resolution of $512 \times 512 \times 31$, where 512×512 represents the spatial resolution and 31 denotes the number of spectral bands. Ten representative scenes are selected from the CAVE dataset, as shown in Fig. 4. The experimental setup for the demosaicing simulations follows the configuration in [17][25], utilizing a 5×5 multispectral filter array. The top-left 500×500 sub-image is cropped, and the first 25 spectral bands are selected from a total of 31. In contrast, the Harvard dataset captures outdoor natural scenes with more diverse and complex spectral and spatial characteristics. The spectral image resolution is $1040 \times 1392 \times 31$. Each spectral band is then resized to 500×500 , and the first 25 bands are selected from the 31 available bands to maintain consistency with the 5×5 mosaic

TABLE I: Comparison of demosaicing results (mean $\pm$ standard deviation) on the 10 CAVE testing scenes by different methods. The PSNR in dB, SSIM, SAM, and ERGAS are reported. A 5×5 binary sub-sampling operator is employed (the best results are highlighted in boldface and the second-best results are labeled with an underline).

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
ItSD	23.32 $\pm$ 1.66	18.32 $\pm$ 2.66	23.77 $\pm$ 1.42	24.65 $\pm$ 2.61	20.92 $\pm$ 3.76	22.09 $\pm$ 2.45	24.08 $\pm$ 3.77	22.65 $\pm$ 2.24	25.54 $\pm$ 3.55	24.73 $\pm$ 1.85	23.00
BTES	35.43 $\pm$ 1.86	24.25 $\pm$ 2.21	34.43 $\pm$ 1.48	25.90 $\pm$ 3.87	35.63 $\pm$ 4.49	23.58 $\pm$ 2.12	30.00 $\pm$ 3.22	28.12 $\pm$ 2.57	32.74 $\pm$ 2.85	28.41 $\pm$ 3.10	29.85
WB	35.57 $\pm$ 1.93	24.40 $\pm$ 2.23	34.41 $\pm$ 1.61	25.94 $\pm$ 4.00	35.77 $\pm$ 4.55	23.53 $\pm$ 2.17	29.89 $\pm$ 3.43	28.27 $\pm$ 2.59	32.85 $\pm$ 2.98	28.45 $\pm$ 3.23	29.91
GRMR	35.47 $\pm$ 1.78	24.46 $\pm$ 2.21	34.23 $\pm$ 1.60	26.10 $\pm$ 3.95	35.65 $\pm$ 4.53	23.69 $\pm$ 2.27	29.88 $\pm$ 3.38	28.43 $\pm$ 2.59	32.95 $\pm$ 2.90	28.32 $\pm$ 3.21	29.92
GAP-TV	37.10 $\pm$ 3.07	23.30 $\pm$ 2.12	34.90 $\pm$ 1.01	27.22 $\pm$ 3.55	37.68 $\pm$ 4.62	23.66 $\pm$ 2.08	34.20 $\pm$ 2.51	27.91 $\pm$ 2.49	34.02 $\pm$ 2.80	28.98 $\pm$ 2.80	30.90
PPID	37.94 $\pm$ 1.64	25.44 $\pm$ 2.43	36.09 $\pm$ 1.08	28.16 $\pm$ 3.16	37.85 $\pm$ 4.76	24.64 $\pm$ 2.41	35.88 $\pm$ 2.40	29.69 $\pm$ 2.43	36.01 $\pm$ 2.72	29.57 $\pm$ 3.85	32.13
MCAN	36.86 $\pm$ 0.61	25.68 $\pm$ 1.31	33.65 $\pm$ 1.08	29.88 $\pm$ 0.37	34.97 $\pm$ 2.42	30.59 $\pm$ 1.05	34.02 $\pm$ 1.02	<u>33.19</u> $\pm$ 0.95	36.99 $\pm$ 1.85	<u>32.01</u> $\pm$ 0.71	32.78
DSP	<u>42.46</u> $\pm$ 0.90	<u>26.43</u> $\pm$ 1.21	<u>37.84</u> $\pm$ 1.10	<u>32.74</u> $\pm$ 2.77	43.59 $\pm$ 4.08	27.07 $\pm$ 1.62	<u>38.81</u> $\pm$ 2.54	32.58 $\pm$ 1.50	40.72 $\pm$ 2.00	31.27 $\pm$ 3.07	<u>35.35</u>
UnNull	44.12 $\pm$ 1.00	30.41 $\pm$ 1.48	37.95 $\pm$ 1.32	33.53 $\pm$ 1.42	<u>43.40</u> $\pm$ 2.75	29.12 $\pm$ 2.03	41.51 $\pm$ 0.93	36.96 $\pm$ 1.86	<u>40.32</u> $\pm$ 2.50	32.02 $\pm$ 1.08	36.93
	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM
ItSD	0.948 $\pm$ 0.03	0.719 $\pm$ 0.14	0.750 $\pm$ 0.06	0.908 $\pm$ 0.03	0.698 $\pm$ 0.07	0.865 $\pm$ 0.06	0.952 $\pm$ 0.02	0.941 $\pm$ 0.02	0.854 $\pm$ 0.06	0.885 $\pm$ 0.05	0.852
BTES	0.988 $\pm$ 0.01	0.933 $\pm$ 0.03	0.976 $\pm$ 0.03	0.891 $\pm$ 0.11	0.965 $\pm$ 0.02	0.927 $\pm$ 0.01	0.927 $\pm$ 0.08	0.956 $\pm$ 0.01	0.920 $\pm$ 0.11	0.926 $\pm$ 0.03	0.941
WB	0.987 $\pm$ 0.01	0.938 $\pm$ 0.03	0.973 $\pm$ 0.03	0.888 $\pm$ 0.12	0.967 $\pm$ 0.02	0.929 $\pm$ 0.01	0.925 $\pm$ 0.08	0.957 $\pm$ 0.01	0.917 $\pm$ 0.12	0.927 $\pm$ 0.03	0.941
GRMR	0.984 $\pm$ 0.01	0.947 $\pm$ 0.02	0.969 $\pm$ 0.03	0.886 $\pm$ 0.12	0.968 $\pm$ 0.02	0.930 $\pm$ 0.01	0.924 $\pm$ 0.08	0.957 $\pm$ 0.01	0.916 $\pm$ 0.12	0.920 $\pm$ 0.03	0.940
GAP-TV	0.995 $\pm$ 0.02	0.905 $\pm$ 0.05	0.989 $\pm$ 0.01	0.948 $\pm$ 0.02	0.983 $\pm$ 0.01	0.925 $\pm$ 0.01	0.974 $\pm$ 0.01	0.955 $\pm$ 0.01	0.982 $\pm$ 0.00	0.936 $\pm$ 0.02	0.959
PPID	0.997 $\pm$ 0.01	<u>0.951</u> $\pm$ 0.03	<u>0.993</u> $\pm$ 0.00	0.965 $\pm$ 0.01	0.980 $\pm$ 0.01	0.951 $\pm$ 0.01	0.985 $\pm$ 0.01	<u>0.976</u> $\pm$ 0.01	0.987 $\pm$ 0.01	0.943 $\pm$ 0.02	<u>0.973</u>
MCAN	0.977 $\pm$ 0.01	0.928 $\pm$ 0.02	0.956 $\pm$ 0.01	0.961 $\pm$ 0.01	0.776 $\pm$ 0.06	<u>0.957</u> $\pm$ 0.01	0.962 $\pm$ 0.01	0.955 $\pm$ 0.01	0.953 $\pm$ 0.01	0.921 $\pm$ 0.01	0.935
DSP	<u>0.998</u> $\pm$ 0.00	0.886 $\pm$ 0.07	0.995 $\pm$ 0.05	<u>0.985</u> $\pm$ 0.03	<u>0.993</u> $\pm$ 0.03	0.938 $\pm$ 0.01	<u>0.989</u> $\pm$ 0.04	0.974 $\pm$ 0.01	<u>0.992</u> $\pm$ 0.04	<u>0.954</u> $\pm$ 0.02	0.970
UnNull	0.999 $\pm$ 0.00	0.989 $\pm$ 0.00	0.995 $\pm$ 0.00	0.992 $\pm$ 0.02	0.994 $\pm$ 0.00	0.984 $\pm$ 0.02	0.997 $\pm$ 0.00	0.994 $\pm$ 0.00	0.994 $\pm$ 0.03	0.975 $\pm$ 0.01	0.991
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
ItSD	0.444 $\pm$ 0.19	0.662 $\pm$ 0.18	0.605 $\pm$ 0.28	0.335 $\pm$ 0.22	0.815 $\pm$ 0.22	0.498 $\pm$ 0.24	0.218 $\pm$ 0.22	0.554 $\pm$ 0.27	0.662 $\pm$ 0.30	0.459 $\pm$ 0.23	0.525
BTES	0.070 $\pm$ 0.15	0.219 $\pm$ 0.19	0.116 $\pm$ 0.14	0.164 $\pm$ 0.17	0.191 $\pm$ 0.21	0.202 $\pm$ 0.15	0.046 $\pm$ 0.12	0.136 $\pm$ 0.18	0.187 $\pm$ 0.18	0.173 $\pm$ 0.15	0.150
WB	0.070 $\pm$ 0.16	0.220 $\pm$ 0.19	0.120 $\pm$ 0.14	0.167 $\pm$ 0.17	0.169 $\pm$ 0.19	0.209 $\pm$ 0.15	0.047 $\pm$ 0.22	0.138 $\pm$ 0.18	0.176 $\pm$ 0.17	0.179 $\pm$ 0.15	0.150
GRMR	0.073 $\pm$ 0.14	0.228 $\pm$ 0.20	0.121 $\pm$ 0.14	0.171 $\pm$ 0.17	0.171 $\pm$ 0.19	0.214 $\pm$ 0.15	0.049 $\pm$ 0.13	0.143 $\pm$ 0.18	0.182 $\pm$ 0.17	0.184 $\pm$ 0.15	0.154
GAP-TV	0.073 $\pm$ 0.11	0.193 $\pm$ 0.17	0.127 $\pm$ 0.11	0.137 $\pm$ 0.15	<u>0.155</u> $\pm$ 0.16	0.163 $\pm$ 0.13	<u>0.029</u> $\pm$ 0.09	0.107 $\pm$ 0.14	0.161 $\pm$ 0.12	0.148 $\pm$ 0.15	0.129
PPID	0.066 $\pm$ 0.14	0.219 $\pm$ 0.20	<u>0.114</u> $\pm$ 0.13	0.158 $\pm$ 0.16	0.169 $\pm$ 0.20	0.193 $\pm$ 0.14	0.037 $\pm$ 0.10	0.137 $\pm$ 0.18	0.175 $\pm$ 0.18	0.174 $\pm$ 0.15	0.144
MCAN	0.116 $\pm$ 0.15	0.286 $\pm$ 0.20	0.179 $\pm$ 0.15	0.200 $\pm$ 0.18	0.387 $\pm$ 0.35	0.213 $\pm$ 0.14	0.055 $\pm$ 0.10	0.180 $\pm$ 0.16	0.391 $\pm$ 0.36	0.207 $\pm$ 0.16	0.221
DSP	<u>0.065</u> $\pm$ 0.10	<u>0.149</u> $\pm$ 0.13	0.124 $\pm$ 0.11	<u>0.130</u> $\pm$ 0.12	0.158 $\pm$ 0.16	0.143 $\pm$ 0.09	0.046 $\pm$ 0.09	<u>0.096</u> $\pm$ 0.14	<u>0.160</u> $\pm$ 0.16	<u>0.147</u> $\pm$ 0.11	0.122
UnNull	0.048 $\pm$ 0.11	0.143 $\pm$ 0.14	0.113 $\pm$ 0.12	0.125 $\pm$ 0.14	0.151 $\pm$ 0.17	<u>0.154</u> $\pm$ 0.10	0.026 $\pm$ 0.08	0.095 $\pm$ 0.13	0.143 $\pm$ 0.15	0.132 $\pm$ 0.12	0.113
	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS
ItSD	47.79	119.3	94.86	52.74	242.9	56.40	26.14	66.23	89.62	45.98	84.20
BTES	12.02	60.90	29.52	34.61	38.47	49.92	12.23	33.81	39.31	31.16	34.20
WB	11.88	60.22	29.42	34.70	37.96	50.27	12.60	33.27	39.37	31.11	34.08
GRMR	12.12	60.30	30.13	34.35	38.40	49.72	12.69	33.32	39.27	31.73	34.20
GAP-TV	09.83	66.65	27.09	29.18	29.79	49.61	07.02	34.51	32.31	29.03	31.50
PPID	08.87	54.22	24.03	19.88	29.52	44.47	05.78	28.16	25.47	28.05	26.85
MCAN	10.27	58.55	34.11	21.12	53.69	35.89	07.58	<u>19.49</u>	27.18	24.41	29.22
DSP	<u>05.28</u>	<u>50.09</u>	18.68	<u>16.00</u>	<u>19.06</u>	<u>35.80</u>	<u>04.66</u>	20.13	<u>17.13</u>	<u>24.18</u>	<u>21.10</u>
UnNull	03.72	26.74	<u>18.83</u>	14.01	14.79	21.77	02.89	12.17	15.59	17.84	14.83

sensor. Similarly, ten representative scenes from the Harvard dataset are chosen for evaluation, as illustrated in Fig. 4. The inclusion of both CAVE and Harvard datasets allows us to evaluate the proposed spectral image reconstruction algorithms under controlled indoor laboratory conditions and challenging outdoor natural scenarios.

Metric. This paper employs four metrics to evaluate spectral image reconstruction performance: the peak signal-to-noise ratio (PSNR), the structural similarity index measure (SSIM) [62], the Spectral Angle Mapper (SAM), and Error Relative Global Dimensionless Synthesis (ERGAS) [63]. PSNR and SSIM are utilized to quantify the spatial fidelity of the data, while SAM is specifically used to reflect the spectral fidelity of the reconstructed spectral images. PSNR and SSIM are computed by averaging across all spectral bands, and SAM is calculated by averaging across all pixels. R3C3: The mean val-

ues with standard deviation are reported. ERGAS is a metric that measures the global error between the reconstructed image and the reference image. Higher PSNR, SSIM, and lower SAM and ERGAS values indicate superior restoration quality.

A. Comparison with Existing Methods

The hyperparameter study of UnNull method will be presented in Sec. IV.B and the implementation code is available ¹. We compare our proposed UnNull algorithm with four traditional algorithms, two hand-prior regularized iterative optimization methods, and two deep learning-based methods. The traditional algorithms considered are Weighted Bilinear (WB) interpolation [16], Iterative Spectral Difference (ItSD) [64], Binary Tree-Based Edge-Sensing (BTES) [13], and Pseudo-Panchromatic Image Difference (PPID) [15] based methods.

¹The code of our UnNull method

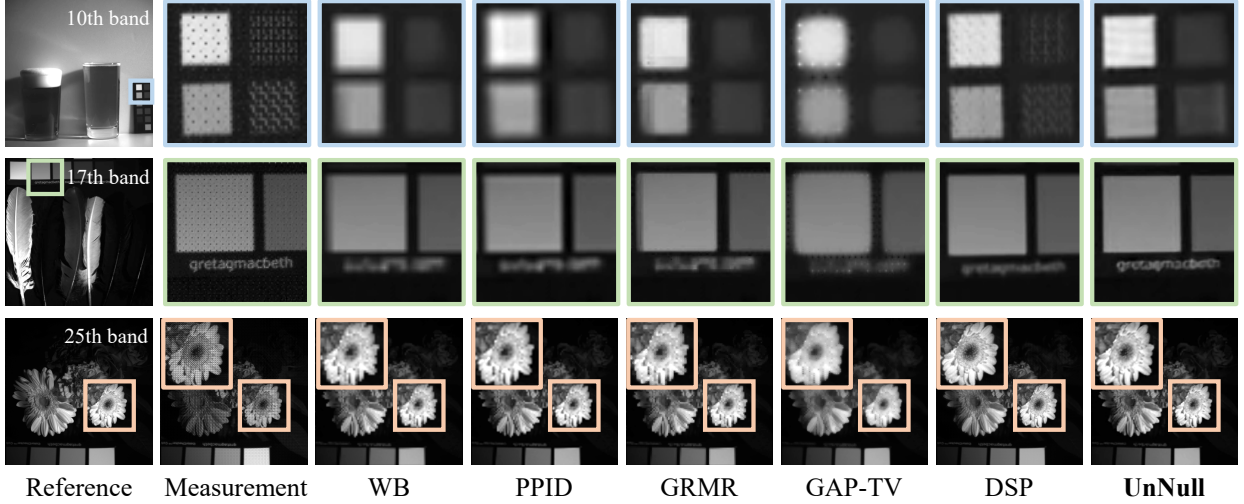


Fig. 5: Demosaiced spectral images of the 10th, 17th, and 25th bands of Scene 7, Scene 8, and Scene 9 from the CAVE dataset by our UnNull method and compared methods.

The model (hand-crafted prior) based methods include Graph and Rank Regularized Matrix Recovery (GRMR) [17] and Generalized Alternating Projection based Total Variation minimization (GAP-TV) [20] based spectral image demosaicing. Additionally, one unsupervised learning-based method (the iterative optimization algorithm with Deep Spectral image Prior (DSP) [25]) and one supervised learning-based method (Mosaic Convolution-Attention Network [32]) are compared. All experiments are conducted on an Intel Core i5-12600K CPU and NVIDIA RTX3080 GPU. The details about the implementation of the compared methods are as follows.

WB is directly applied using the standard bilinear interpolation formula. Missing values were estimated by considering neighboring pixels within a $(2L - 1) \times (2L - 1)$ window.

ItSD algorithm reconstructs the spectral image by iteratively minimizing spectral differences between the measured mosaic and the interpolated image. The number of iterations is set to 5. In each iteration, a smoothing filter H (a 7×7 Gaussian kernel) is used to convolve the spectral differences. The final spectral images are obtained by adding the mean of the interpolated image to the smoothed spectral differences.

BTES relies on a dominant spectral band (e.g., the G band), which is unavailable in our case. Therefore, we adapt this approach by defining a sequence of interpolation steps. This sequence simulates the edge-sensing mechanism by performing circular shifts of the filter pattern for each spectral band, enabling the algorithm to adaptively infer missing spectral information for each pixel. Specifically, for each spectral band, the interpolation process involves a progressive procedure, with each step calculating the value of missing pixels based on a weighted sum of neighboring pixels.

PPID algorithm operates by leveraging the difference between the raw mosaic image and a Pseudo-Panchromatic Image (PPI). The PPI is computed as the average of weighted bilinear interpolated images. The difference is then smoothed using a $(2L - 1) \times (2L - 1)$ Gaussian kernel. Finally, the reconstructed spectral images are obtained by summing the smoothed differences and the PPI.

GRMR method is implemented according to the original paper. The image patch size is set to 5×5 , and the number of iterations is set to 20. The low-rank prior is enforced using Singular Value Decomposition (SVD), with the rank set to 2. In conclusion, the WB, ItSD, BTES, PPID, and GRMR methods are implemented using the code available at ².

GAP-TV algorithm is implemented using an iterative method for solving the main variable and minimizing the total variation. The number of iterations is set to 600. In each iteration, the main variable is updated through alternating projections, and the total variation (TV) minimization step is performed to suppress noise and preserve sharp edges. The TV minimization is executed using a splitting and proximal gradient-based approach. Hyperparameters are optimized for optimal performance.

DSP method uses a Plug-and-Play (PnP) framework, where the spectral image prior is represented by an untrained neural network. The total number of iterations for ADMM-DSP is set to 40, with each iteration involving training the network with 400 epochs. The network architecture follows the design used in the original paper. Balancing factors and the learning rate are optimized to achieve the best performance.

MCAN is a supervised learning-based method. The training dataset consists of 21 scenes from the CAVE dataset, with the last 10 CAVE scenes and 10 selected Harvard scenes used for testing. The network architecture is consistent with that in the original work, while the relative position matrix is adjusted to match the 5×5 mosaic sensor. The method is trained with a total of 10,000 epochs, using L_1 Charbonnier loss. Implementation of MCAN is based on their code at ³.

Results of spectral image demosaicing on the CAVE dataset. Firstly, we provide the results of the demosaicing spectral image with a 5×5 filter array. This task aims to reconstruct 25 spectral images from the raw image. Table I provides a comprehensive performance comparison of existing spectral

²The code of compared methods

³The code of MCAN methods

TABLE II: Comparison of demosaicing results (mean $\pm$ standard deviation) on the 10 Harvard testing scenes by different methods. The PSNR in dB, SSIM, SAM, and ERGAS are reported. A 5×5 binary sub-sampling operator is employed (the best results are highlighted in boldface and the second-best results are labeled with an underline).

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
ItSD	20.00 $\pm$ 4.38	26.75 $\pm$ 4.49	22.52 $\pm$ 3.93	22.91 $\pm$ 3.31	21.44 $\pm$ 4.46	25.29 $\pm$ 2.29	22.24 $\pm$ 3.90	28.35 $\pm$ 3.53	23.56 $\pm$ 4.33	23.70 $\pm$ 4.22	23.68
WB	28.23 $\pm$ 6.58	33.64 $\pm$ 5.90	32.10 $\pm$ 7.54	29.55 $\pm$ 7.02	30.20 $\pm$ 6.26	29.91 $\pm$ 5.25	31.69 $\pm$ 7.72	36.20 $\pm$ 6.90	31.28 $\pm$ 7.38	30.58 $\pm$ 5.91	31.34
BTES	28.24 $\pm$ 6.48	33.79 $\pm$ 5.78	32.29 $\pm$ 7.47	29.57 $\pm$ 6.98	30.34 $\pm$ 6.19	29.94 $\pm$ 5.25	31.68 $\pm$ 7.64	36.23 $\pm$ 6.85	31.39 $\pm$ 7.31	30.66 $\pm$ 5.76	31.41
GRMR	28.34 $\pm$ 6.63	33.72 $\pm$ 5.93	32.26 $\pm$ 7.55	29.69 $\pm$ 7.06	30.23 $\pm$ 6.26	30.03 $\pm$ 2.28	31.75 $\pm$ 7.73	36.24 $\pm$ 6.94	31.35 $\pm$ 7.40	30.75 $\pm$ 5.94	31.44
MCAN	27.44 $\pm$ 1.35	33.55 $\pm$ 1.57	32.09 $\pm$ 2.96	30.69 $\pm$ 2.58	31.09 $\pm$ 1.53	32.99 $\pm$ 1.77	31.59 $\pm$ 2.21	37.13 $\pm$ 2.59	31.94 $\pm$ 2.18	30.68 $\pm$ 1.44	31.92
PPID	29.99 $\pm$ 3.36	35.48 $\pm$ 3.41	32.87 $\pm$ 5.09	30.33 $\pm$ 5.24	32.89 $\pm$ 3.30	32.57 $\pm$ 3.85	32.17 $\pm$ 4.56	37.06 $\pm$ 4.57	32.42 $\pm$ 4.01	34.34 $\pm$ 2.96	33.01
GAP-TV	29.07 $\pm$ 6.22	35.78 $\pm$ 5.76	33.64 $\pm$ 7.60	29.73 $\pm$ 6.95	33.18 $\pm$ 6.00	30.85 $\pm$ 5.51	33.08 $\pm$ 7.80	37.46 $\pm$ 7.06	33.58 $\pm$ 7.27	35.21 $\pm$ 5.90	33.16
DSP	<u>32.32$\pm$3.31</u>	<u>36.82$\pm$3.65</u>	<u>35.36$\pm$5.38</u>	<u>31.23$\pm$5.13</u>	<u>34.96$\pm$3.34</u>	32.51 $\pm$ 3.76	<u>33.78$\pm$5.87</u>	<u>37.96$\pm$5.74</u>	<u>35.44$\pm$6.34</u>	<u>38.82$\pm$5.38</u>	<u>34.92</u>
UnNull	36.94$\pm$4.45	38.82$\pm$5.94	37.22$\pm$7.09	37.18$\pm$5.73	38.71$\pm$4.80	41.86$\pm$4.61	39.58$\pm$6.44	42.94$\pm$6.35	38.84$\pm$6.12	44.13$\pm$4.53	39.62
	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM
	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM
ItSD	<u>0.961$\pm$0.01</u>	0.968 $\pm$ 0.01	0.913 $\pm$ 0.02	0.906 $\pm$ 0.06	0.962 $\pm$ 0.01	0.889 $\pm$ 0.06	<u>0.941$\pm$0.01</u>	0.906 $\pm$ 0.04	<u>0.899$\pm$0.00</u>	0.969 $\pm$ 0.01	0.931
WB	0.896 $\pm$ 0.11	0.959 $\pm$ 0.02	<u>0.921$\pm$0.01</u>	0.945 $\pm$ 0.02	0.895 $\pm$ 0.12	0.841 $\pm$ 0.11	0.894 $\pm$ 0.07	<u>0.940$\pm$0.08</u>	0.813 $\pm$ 0.09	0.925 $\pm$ 0.11	0.893
BTES	0.900 $\pm$ 0.10	0.960 $\pm$ 0.02	0.918 $\pm$ 0.01	0.944 $\pm$ 0.03	0.900 $\pm$ 0.11	0.830 $\pm$ 0.11	0.896 $\pm$ 0.06	0.936 $\pm$ 0.10	0.814 $\pm$ 0.08	0.926 $\pm$ 0.10	0.892
GRMR	0.893 $\pm$ 0.11	0.957 $\pm$ 0.02	0.919 $\pm$ 0.01	0.944 $\pm$ 0.02	0.893 $\pm$ 0.11	0.846 $\pm$ 0.11	0.894 $\pm$ 0.07	0.939 $\pm$ 0.08	0.820 $\pm$ 0.09	0.923 $\pm$ 0.11	0.893
MCAN	0.763 $\pm$ 0.15	0.778 $\pm$ 0.17	0.791 $\pm$ 0.19	0.909 $\pm$ 0.09	0.785 $\pm$ 0.19	0.822 $\pm$ 0.13	0.753 $\pm$ 0.22	0.856 $\pm$ 0.13	0.683 $\pm$ 0.22	0.763 $\pm$ 0.12	0.790
PPID	0.917 $\pm$ 0.13	<u>0.975$\pm$0.03</u>	0.916 $\pm$ 0.09	<u>0.963$\pm$0.03</u>	0.946 $\pm$ 0.11	<u>0.915$\pm$0.09</u>	0.878 $\pm$ 0.17	0.937 $\pm$ 0.05	0.844 $\pm$ 0.16	0.959 $\pm$ 0.09	0.925
GAP-TV	0.952 $\pm$ 0.00	<u>0.975$\pm$0.00</u>	0.921 $\pm$ 0.01	0.916 $\pm$ 0.04	<u>0.972$\pm$0.01</u>	0.897 $\pm$ 0.04	0.948 $\pm$ 0.05	0.929 $\pm$ 0.08	0.884 $\pm$ 0.06	<u>0.991$\pm$0.01</u>	<u>0.949</u>
DSP	0.936 $\pm$ 0.06	0.914 $\pm$ 0.12	0.866 $\pm$ 0.10	0.889 $\pm$ 0.07	0.918 $\pm$ 0.07	0.892 $\pm$ 0.11	0.846 $\pm$ 0.13	0.846 $\pm$ 0.16	0.843 $\pm$ 0.20	0.958 $\pm$ 0.06	0.901
UnNull	0.963$\pm$0.04	0.979$\pm$0.01	0.958$\pm$0.01	0.985$\pm$0.02	0.985$\pm$0.01	0.995$\pm$0.00	0.981$\pm$0.01	0.986$\pm$0.00	0.979$\pm$0.01	0.997$\pm$0.00	0.981
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
ItSD	0.422 $\pm$ 0.01	0.369 $\pm$ 0.08	0.534 $\pm$ 0.08	0.496 $\pm$ 0.08	0.438 $\pm$ 0.09	0.471 $\pm$ 0.11	0.491 $\pm$ 0.10	0.542 $\pm$ 0.10	0.460 $\pm$ 0.07	0.382 $\pm$ 0.10	0.461
WB	0.115 $\pm$ 0.12	0.083 $\pm$ 0.09	0.114 $\pm$ 0.10	0.127 $\pm$ 0.12	0.091 $\pm$ 0.11	0.129 $\pm$ 0.15	0.126 $\pm$ 0.13	0.128 $\pm$ 0.11	0.125 $\pm$ 0.13	0.104 $\pm$ 0.12	0.114
BTES	0.109 $\pm$ 0.11	0.079 $\pm$ 0.09	0.108 $\pm$ 0.10	0.120 $\pm$ 0.11	0.087 $\pm$ 0.11	0.123 $\pm$ 0.14	0.121 $\pm$ 0.12	0.122 $\pm$ 0.11	0.119 $\pm$ 0.12	0.101 $\pm$ 0.12	0.109
GRMR	0.120 $\pm$ 0.12	0.085 $\pm$ 0.09	0.119 $\pm$ 0.10	0.132 $\pm$ 0.12	0.097 $\pm$ 0.12	0.135 $\pm$ 0.15	0.131 $\pm$ 0.14	0.131 $\pm$ 0.11	0.129 $\pm$ 0.13	0.106 $\pm$ 0.13	0.119
MCAN	0.132 $\pm$ 0.08	0.103 $\pm$ 0.04	0.118 $\pm$ 0.06	0.126 $\pm$ 0.09	0.104 $\pm$ 0.07	0.142 $\pm$ 0.14	0.136 $\pm$ 0.10	0.122 $\pm$ 0.07	0.130 $\pm$ 0.10	0.127 $\pm$ 0.09	0.124
PPID	0.103 $\pm$ 0.09	0.071 $\pm$ 0.05	0.103 $\pm$ 0.07	0.116 $\pm$ 0.09	0.080 $\pm$ 0.08	0.125 $\pm$ 0.15	0.117 $\pm$ 0.11	0.116 $\pm$ 0.08	0.115 $\pm$ 0.11	0.097 $\pm$ 0.11	0.104
GAP-TV	0.074 $\pm$ 0.06	0.050$\pm$0.03	0.068$\pm$0.04	<u>0.084$\pm$0.07</u>	<u>0.055$\pm$0.05</u>	<u>0.081$\pm$0.09</u>	<u>0.076$\pm$0.08</u>	<u>0.085$\pm$0.06</u>	0.076$\pm$0.07	0.066$\pm$0.06	<u>0.072</u>
DSP	0.080 $\pm$ 0.05	0.072 $\pm$ 0.03	0.099 $\pm$ 0.04	0.119 $\pm$ 0.04	0.070 $\pm$ 0.04	0.143 $\pm$ 0.07	0.107 $\pm$ 0.08	0.132 $\pm$ 0.04	0.084 $\pm$ 0.05	0.081 $\pm$ 0.05	0.099
UnNull	0.071$\pm$0.05	<u>0.062$\pm$0.03</u>	<u>0.076$\pm$0.04</u>	0.075$\pm$0.05	0.050$\pm$0.04	0.074$\pm$0.08	0.069$\pm$0.06	0.083$\pm$0.05	<u>0.078$\pm$0.06</u>	<u>0.069$\pm$0.06</u>	0.071
	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS
	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS
ItSD	256.2	140.6	340.9	316.1	205.5	278.7	351.6	308.0	278.4	152.0	262.8
WB	22.49	17.75	26.72	40.25	20.84	53.50	25.07	34.19	23.74	21.81	28.64
BTES	22.28	17.36	26.39	40.12	20.31	52.67	25.03	34.03	23.36	21.28	28.28
GRMR	22.18	17.58	26.51	39.57	20.74	52.72	24.88	34.00	23.54	21.38	29.31
MCAN	50.83	30.09	50.59	67.22	32.40	66.08	58.39	55.37	43.89	35.32	49.02
PPID	22.50	17.24	29.97	42.11	18.58	<u>42.58</u>	32.11	36.81	28.26	15.51	28.57
GAP-TV	19.74	<u>15.55</u>	<u>22.48</u>	<u>39.33</u>	<u>14.06</u>	<u>45.78</u>	<u>21.20</u>	<u>29.45</u>	<u>17.84</u>	<u>12.49</u>	<u>23.79</u>
DSP	<u>15.53</u>	19.25	34.66	54.45	20.69	92.01	38.83	75.08	31.09	12.86	39.45
UnNull	08.80	09.60	15.08	17.19	07.60	13.03	10.19	14.66	09.83	04.37	11.04

image demosaicing algorithms on the 10 CAFE testing scenes. Our proposed method (UnNull) achieves the best performance in terms of the average PSNR, SSIM, SAM, and ERGAS. For example, UnNull achieves an average PSNR of 36.93 dB and an average SSIM of 0.991, outperforming all existing methods. Compared with the previously best-performing algorithm (DSP), UnNull demonstrates a significant improvement of 1.58 dB in average PSNR. Regarding average ERGAS, our algorithm exhibits a 12.75 improvement compared to the previous best-performing algorithm (GAP-TV).

For visualizing the demosaiced spectral image with a 5×5 binary sub-sampling operator, Fig. 5 displays three scenes generated by six different methods. The 10th, 17th, and 25th bands are selected for comparison, respectively. From the figure, we can see that the traditional methods (*e.g.*, WB, PPID, and GRMR) present blurred images, while the mosaic

patterns are prevalent in the image reconstructed by GAP-TV and DSP. Our algorithm produces spectral images with distinct textures and exceptional fidelity compared to other methods.

Results of spectral image demosaicing on the Harvard dataset. Table II summarizes the demosaicing results on the 10 Harvard testing scenes, with four key evaluation metrics (PSNR, SSIM, SAM, and ERGAS). The results demonstrate that our method achieves consistently high PSNR and SSIM values across all scenes, significantly outperforming other methods. For instance, on scene 4, our method attains a PSNR of 37.18 dB, compared to 29.69 dB by GRMR and 31.23 dB by DSP. Similarly, in terms of SAM, which evaluates the spectral fidelity, our method achieves the lowest average SAM of 0.071, indicating the most accurate spectral reconstruction among all tested methods. The proposed method also yields the lowest ERGAS values across the dataset, with an average

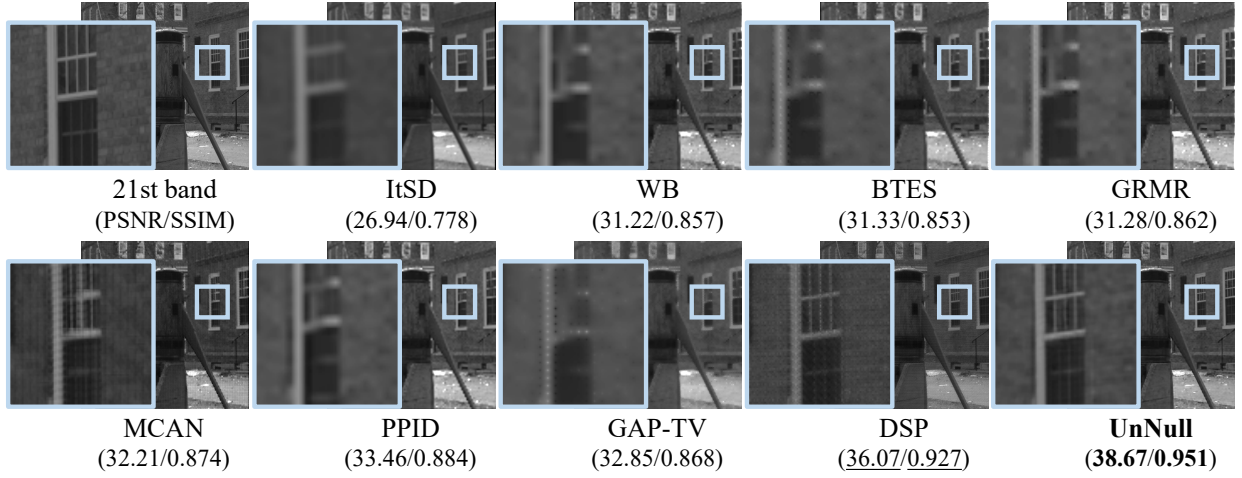


Fig. 6: Comparison of the 21st band of Scene 8 from the Harvard dataset demosaiced by our methods and compared algorithms. The PSNR and SSIM of the 21st band are presented, accordingly.

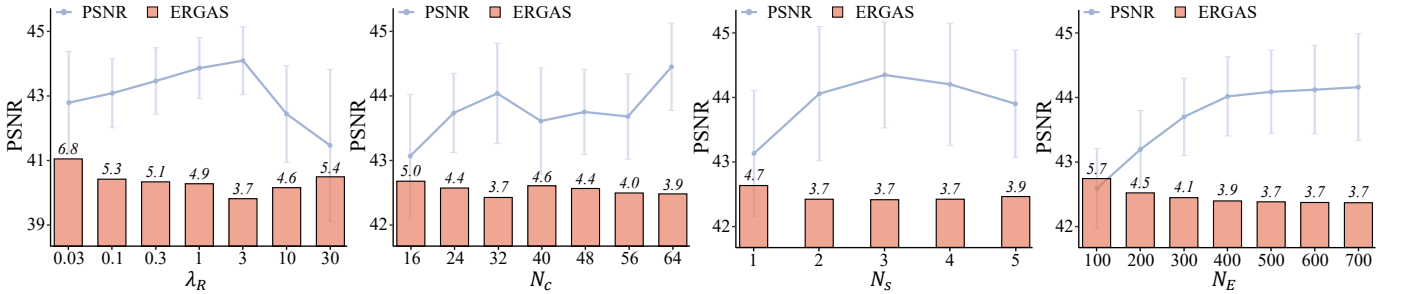


Fig. 7: The effect of balancing factor λ_R , the number of feature maps N_c , the number of network stages N_S , and the number of training epochs N_E . Spectral image demosaicing experiments are conducted on Scene 1 from the CAVE dataset.

of 11.04, highlighting its robust performance in producing globally accurate reconstructions. In conclusion, the results on the Harvard dataset not only reinforce the effectiveness of our method but also demonstrate its strong generalization capability.

Moreover, to provide visual reconstruction results on the Harvard dataset, we present the reconstructed spectral image of Scene 8. In Fig. 6, the 21st band is selected for comparison. The figure clearly demonstrates that our proposed UnNull algorithm produces finer image details and more high-fidelity spectral images. Additionally, we include the results of PSNR and SSIM values of each method corresponding to the selected bands. The highest PSNR and SSIM values also underscore the superior reconstruction accuracy of our method.

B. Ablation Study

Effect of hyperparameters. To investigate the sensitivity of the proposed UnNull method to various hyperparameters, we conducted a series of experiments, as summarized in Fig. 7. These experiments evaluate the influence of the balancing factor λ_R , the number of feature map channels N_c , the number of network stages N_S , and the number of training epochs N_E , using Scene 1 from the CAVE dataset with a 5×5 sub-sampling operator. The parameter λ_R is varied from 0.03 to 30. The results in Fig. 7 demonstrate that the optimal PSNR and

ERGAS values are achieved when $\lambda_R=3$. Next, the influence of N_c is analyzed over a range from 16 to 64, with increments of 8. The best PSNR is observed when $N_c=64$, while the best ERGAS result is obtained when $N_c=32$. To balance computational efficiency and performance, $N_c=32$ is chosen as the optimal setting, ensuring competitive demosaicing quality with reasonable runtime. Then, the number of network stages N_S is tested, which is varied from 1 to 5. The results show that performance improves with additional stages, peaking at $N_S=3$. Beyond this point, further increases yield diminishing returns. Finally, the training epochs N_E are evaluated, which is varied from 100 to 700 in steps of 100. The results indicate that reconstruction performance improves consistently with more epochs, stabilizing around $N_E=500$. This suggests that a few training epochs are sufficient for our method to achieve high-quality spectral reconstruction without overfitting.

Effect of range-nullspace decomposition strategy. To evaluate the effectiveness of our proposed UnNull framework, we conduct experiments to compare UnNull with these two settings: $\hat{x} = f_\theta(Gy)$ and $\hat{x} = Gy + f_\theta(Gy)$. For the first configuration, the network output is directly treated as the reconstruction spectral images, like the deep image prior [23]. The second configuration can be seen as utilizing the untrained neural network to learn the residual of range subspace. The network architecture $f_\theta(\cdot)$ and loss function maintain the same one used in our UnNull method for fair. Experiments

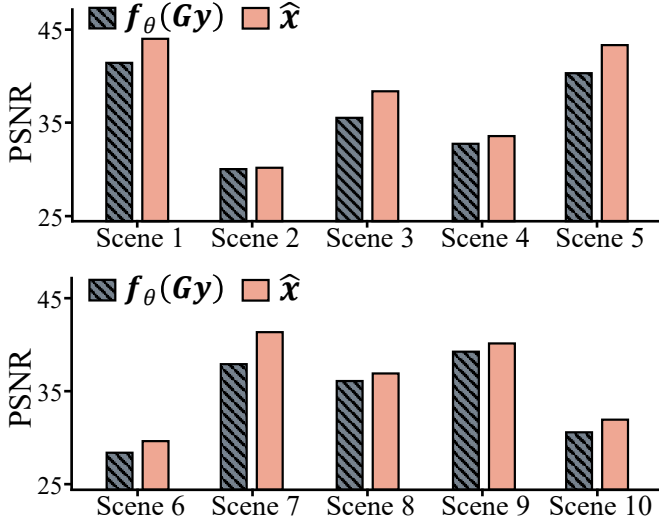


Fig. 8: Comparison of PSNR results between the direct network output $f_{\theta}(Gy)$ and the our UnNull reconstruction $\hat{x}$ on 10 scenes.

TABLE III: Effect of the L_p norm, SC block, CAM block, STV regularization, optimizer, and difference untrained neural network priors. Spectral image demosaicing experiments are conducted on CAVE Scene 1.

Different Settings	PSNR	SSIM	SAM	ERGAS
Ours	44.12 ± 1.00	0.999 ± 0.00	0.048 ± 0.11	03.72
w L_2	42.11 ± 2.17	0.998 ± 0.02	0.066 ± 0.12	05.64
w L_1	44.56 ± 1.04	0.999 ± 0.00	0.059 ± 0.11	03.94
w/o SC	43.13 ± 1.30	0.998 ± 0.00	0.053 ± 0.12	04.30
w/o CAM	42.38 ± 1.09	0.999 ± 0.00	0.056 ± 0.12	05.24
w/o STV	43.07 ± 1.40	0.999 ± 0.00	0.051 ± 0.12	04.41
w SGD	33.43 ± 2.38	0.990 ± 0.00	0.085 ± 0.15	15.43
$Gy + f_{\theta}(Gy)$	38.15 ± 1.69	0.996 ± 0.01	0.069 ± 0.14	08.66
$f_{\theta}(Gy)$	41.48 ± 1.41	0.998 ± 0.03	0.059 ± 0.11	06.02

are performed on the CAVE dataset, utilizing a 5×5 binary sub-sampling operator. Firstly, we compare the PSNR, SSIM, SAM, and ERGAS of Scene 1 reconstructed by UnNull and the two variants in Table III. It can be observed that our proposed method consistently outperforms the two variations. Compared to $Gy + f_{\theta}(Gy)$, the variant $f_{\theta}(Gy)$ achieves better performance but still falls short of our method. Moreover, comparisons of PSNR results for each scene reconstructed by UnNull ($\hat{x}$) and the variant $f_{\theta}(Gy)$ are shown in Fig. 8. These results reveal that relying solely on the network output produces suboptimal performance. This demonstrates the critical role of the range-nullspace decomposition strategy in overcoming the limitation of existing untrained neural network prior-based methods and enhancing the overall performance of spectral image reconstruction.

Effectiveness of the Skip Connection (SC). The Skip Connection (SC) block constitutes a pivotal component within the encoder-decoder network architecture, facilitating the fusion of

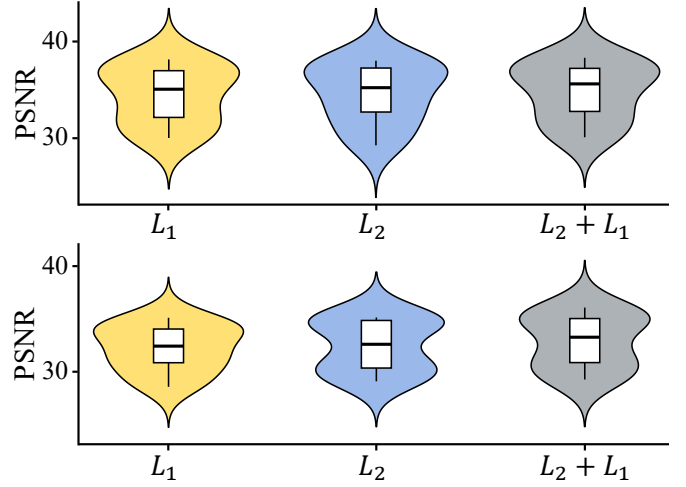


Fig. 9: Comparison of different data fidelity terms under various noise conditions. The top plot shows results for Gaussian noise, while the bottom plot presents results for Gaussian-sparse mixed noise.

features across different hierarchical levels. Its effectiveness is evaluated on Scene 1 with four metrics. As shown in Table III, a notable performance decrease emerges between the network configurations with and without skip connections. Specifically, the network lacking skip connections (denoted as ‘w/o SC’ in the table) yields inferior results compared to its counterpart equipped with the SC block (the first row, denoted as ‘ours’). This observation underscores the significant role played by skip connections in promoting information flow and enhancing feature integration, contributing to improved performance in spectral image demosaicing.

Effectiveness of the Channel Attention Module (CAM). Furthermore, we assess the impact of the Channel Attention Module (CAM) block via experiments conducted on Scene 1. To evaluate its effectiveness, we substitute the CAM block with a standard convolutional layer with a kernel size of 3 and a stride of 1. Table III provides a comparison between the network equipped with the CAM block (represented in the first row) and the network without CAM (denoted as ‘w/o CAM’ in the table). Notably, upon incorporating the CAM block, an improvement in terms of PSNR, SAM, and ERGAS is observed. These results underscore the utility of the CAM in feature extraction and enhancing the overall performance of the network for spectral image reconstruction.

Effect of the neural network optimizer. Table III provides insights into the influence of the employed neural network optimizer on spectral image demosaicing performance. The experiments are conducted on Scene 1 to evaluate the effectiveness of different optimizers. Specifically, we compare the performance of our UnNull model utilizing the ADAM optimizer against that using the Stochastic Gradient Descent (SGD) optimizer. The learning rates of both ADAM and SGD are carefully tuned to ensure optimal performance. As depicted in the table, the model trained with the ADAM optimizer consistently outperforms its counterpart trained with the SGD optimizer across all metrics.

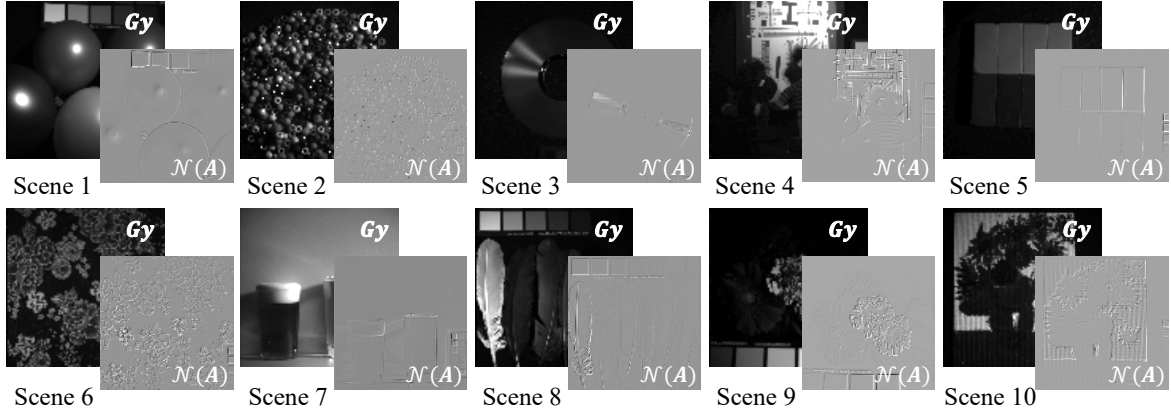


Fig. 10: The 10th band of the spectral images projected onto its range (Gy) and nullspace ($N(A)$), respectively. Features on the learned nullspace are high-frequency information (e.g., edges).

TABLE IV: Evaluation of method robustness on CAVE Scene 1. Measurements are corrupted with Gaussian noise with a level of 0.1 and a Gaussian and sparse mixed noise.

G (0.1)	PSNR	SSIM	SAM	ERGAS
ItSD	23.22±1.69	0.943±0.03	0.444±0.19	48.56
GAP-TV	30.75±0.79	0.961±0.01	0.142±0.14	19.97
GRMR	33.40±1.11	0.979±0.01	0.138±0.17	14.89
WB	33.54±1.15	0.982±0.01	0.135±0.17	14.70
BTES	33.54±1.11	0.983±0.01	0.130±0.17	14.67
DSP	34.35±0.59	0.981±0.01	0.131±0.14	12.97
PPID	<u>35.07±0.67</u>	<u>0.991±0.01</u>	<u>0.129±0.16</u>	<u>12.12</u>
UnNull	37.01±0.24	0.993±0.00	0.122±0.15	09.64
G + S	PSNR	SSIM	SAM	ERGAS
ItSD	23.18±1.67	0.942±0.04	0.444±0.19	48.81
GAP-TV	30.55±0.77	0.965±0.01	<u>0.111±0.14</u>	20.41
GRMR	30.12±0.75	0.971±0.01	0.164±0.21	21.64
WB	30.37±0.77	0.974±0.01	0.164±0.21	21.05
BTES	30.74±0.75	0.977±0.01	0.160±0.20	20.16
DSP	<u>33.63±0.55</u>	0.975±0.01	0.149±0.15	<u>14.31</u>
PPID	31.17±0.60	<u>0.985±0.01</u>	0.157±0.20	19.02
UnNull	34.67±0.68	0.990±0.01	0.131±0.15	12.68

TABLE V: Comparison running time of performing spectral image demosaicing on different datasets and the implementation platform.

Dataset	PPID (CPU)	GRMR (CPU)	GAP-TV (GPU)	DSP (GPU)	UnNull (GPU)
CAVE Data	1.3 s	81.7 s	67.4 s	53.5 s	26.4 s
Noisy Data	1.3 s	81.7 s	40.6 s	63.5 s	18.5 s
Real Data	4.0 s	155.5 s	204.1 s	153.2 s	9.8 s

C. Discussion

Discussion on noise effects and data fidelity terms.

The selection of data fidelity terms should align with the characteristics of the noise distribution ($p(e)$). For example, the L_2 norm, derived from maximum likelihood estimation, is optimal for Gaussian noise, while the L_1/L_0 norms are better suited to handle sparse noise. First, we test the proposed

method under scenarios involving Gaussian noise and mixed Gaussian-sparse noise using three fidelity terms: the L_1 norm, the L_2 norm, and a combination of $L_2 + L_1$ norm. The noise level of Gaussian noise is set to 0.01, and 0.2% pixels are affected by sparse noise. The results, presented in Fig. 9, show utilizing the L_2 norm performs slightly better than the L_1 norm when measurements are corrupted by Gaussian noise or mixed noise. Moreover, results demonstrate the combined $L_2 + L_1$ loss obtains superior robustness across diverse noise scenarios.

Furthermore, under noise-free conditions, we also compared the performance of our method using these three fidelity terms. As shown in Table III, the experiments reveal that the model with L_1 norm can outperform the L_2 norm in all metrics. Moreover, the combined $L_2 + L_1$ norm (the first row, denoted as ‘ours’) achieves the best performance. Finally, considering the noise effects in real scenes and the performance in simulation experiments, we adopted the combined $L_2 + L_1$ norm as the data fidelity term, which can balance robustness and accuracy across diverse scenarios.

Robustness to imaging noise. Here, we evaluate the robustness of our UnNull algorithm and the compared methods. The measurements (raw images) are contaminated with Gaussian noise, and Gaussian-sparse mixed noise, respectively. The Gaussian noise is added with a mean of 0.5 and a noise level of 0.01. For sparse noise, 0.2% of the pixels are affected. The demosaicing results on Scene 1, utilizing a 5×5 binary subsampling operator, are presented in Table IV. Upon analyzing the results presented in the table and in Table I (clean measurements), it is apparent that previous methods suffer from significant performance degradation under the influence of noise corruption. Conversely, the proposed UnNull method exhibits superior robustness, maintaining consistently high performance levels across all evaluated scenes. This robustness underscores the effectiveness of our approach in real-world scenarios where data corruption is commonplace.

Algorithm running time. Convex programming and unsupervised learning-based methods often suffer from slow algorithm running times, making them less practical for real-time applications. For instance, the iterative optimization-based method GRMR requires 81.7 seconds to reconstruct the

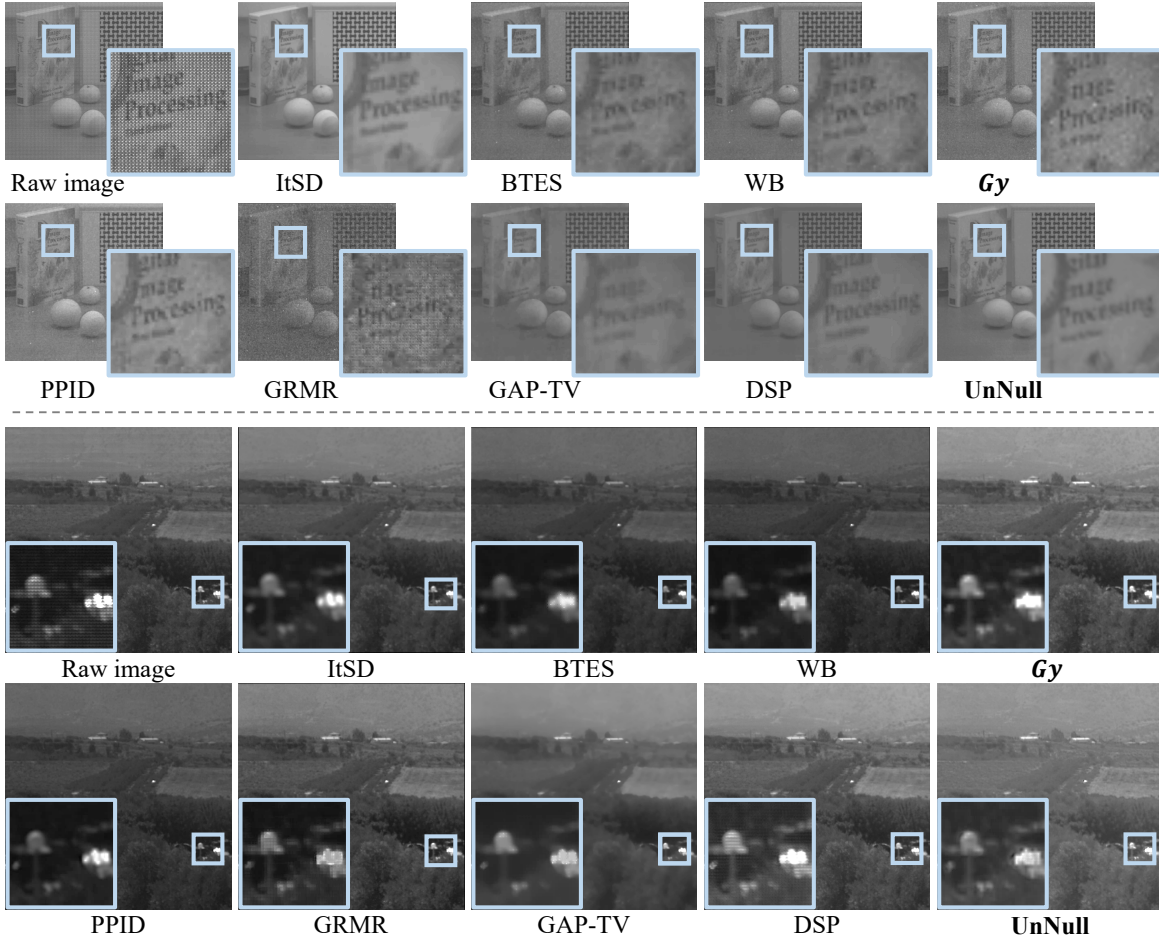


Fig. 11: The raw image from a real 5×5 camera (top) and comparison of the 10th band image generated by our UnNull and compared methods. The raw image from a real 4×4 camera (bottom) and the reconstructed 10th band images.

TABLE VI: Demosaicing results on difference mosaic sensor, where BTES-I and BTES-II are proposed in [13] and [14], respectively.

Mosaic	PSNR	SSIM	SAM	ERGAS
GBRG	31.29 ± 1.43	0.940 ± 0.02	0.109 ± 0.12	21.85
3×3	33.00 ± 2.61	0.976 ± 0.01	0.177 ± 0.11	20.76
4×4	31.44 ± 2.35	0.950 ± 0.02	0.162 ± 0.12	24.14
BTES-I	30.50 ± 0.72	0.936 ± 0.01	0.124 ± 0.13	24.00
BTES-II	30.51 ± 0.98	0.939 ± 0.01	0.132 ± 0.14	24.12

spectral image of a single captured scene with a resolution of 500×500 on the CAVE dataset. Similarly, the DSP method, which integrates unsupervised learning with an optimization framework, takes 53.5 seconds for a single reconstruction task. In contrast, our proposed UnNull algorithm exhibits significantly enhanced computational efficiency across various datasets and scenarios. For example, it achieves a running time of just 26.4 seconds for spectral image reconstruction on the CAVE dataset, as detailed in Table V.

This efficiency is even more pronounced when processing noisy or real datasets. For handling measurements corrupted by the mixed Gaussian and sparse noise, UnNull achieves

a runtime of 18.5 seconds, leveraging fewer training epochs to mitigate overfitting to noisy measurements. Our method demonstrates an obvious advantage over 63.5 seconds for DSP and 40.6 seconds for GAP-TV. For real data with a size of $1000 \times 1000 \times 25$, UnNull reconstructs spectral images in 9.8 seconds using 40 epochs, significantly outperforming GRMR (155.5 seconds), DSP (153.2 seconds), and GAP-TV (204.1 seconds). These results underscore the remarkable computational efficiency of the proposed method, making it well-suited for practical applications requiring fast and accurate spectral image demosaicing.

Visualization of spectral images projected onto the range and nullspace. As previously mentioned, the data projected onto the range predominantly depicts low-frequency information, whereas the portion onto the nullspace captures high-frequency features. In Fig. 10, we present visualizations of the range and nullspace representations of Scene 1 to Scene 10. These experiments utilize a 5×5 binary sub-sampling operator. The visualizations demonstrate the distinct characteristics captured within the range and nullspace projections. Specifically, the data within the nullspace captures clear and desired high-frequency features, such as edges and fine textures. This highlights the effectiveness of our nullspace learning method.

Results on different mosaic patterns. To evaluate the

TABLE VII: Comparison of reconstruction results (mean $\pm$ standard deviation) on the 10 CAVE testing scenes by different methods. The PSNR in dB, SSIM, SAM, and ERGAS are reported. A 5×5 filter response function-based sub-sampling operator is employed (the best results are highlighted in boldface and the second-best results are labeled with an underline).

Method	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	Scene 9	Scene 10	Average
	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR	PSNR
$Gy + f_\theta(Gy)$	36.25 $\pm$ 4.94	27.02 $\pm$ 3.09	29.91 $\pm$ 5.00	30.65 $\pm$ 4.39	37.24 $\pm$ 3.59	27.14 $\pm$ 2.61	35.53 $\pm$ 4.28	31.36 $\pm$ 5.16	34.69 $\pm$ 3.59	29.35 $\pm$ 2.85	31.91
$f_\theta(Gy)$	37.66 $\pm$ 5.81	28.37 $\pm$ 3.57	30.57 $\pm$ 4.45	33.24 $\pm$ 5.30	39.15 $\pm$ 4.92	29.67 $\pm$ 3.10	36.03 $\pm$ 4.47	33.54 $\pm$ 5.93	35.59 $\pm$ 4.79	32.09 $\pm$ 3.12	33.49
UnNull	40.16 $\pm$ 6.34	29.29 $\pm$ 3.7	31.48 $\pm$ 5.59	34.76 $\pm$ 5.66	<u>39.46</u> $\pm$ 4.56	<u>28.64</u> $\pm$ 2.71	38.28 $\pm$ 4.64	34.49 $\pm$ 6.15	<u>38.15</u> $\pm$ 3.97	31.00 $\pm$ 3.11	34.57
	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM	SSIM
$Gy + f_\theta(Gy)$	0.993 $\pm$ 0.01	0.955 $\pm$ 0.08	0.937 $\pm$ 0.08	0.968 $\pm$ 0.05	0.926 $\pm$ 0.12	0.957 $\pm$ 0.08	0.999 $\pm$ 0.02	0.976 $\pm$ 0.03	0.963 $\pm$ 0.07	0.953 $\pm$ 0.05	0.963
$f_\theta(Gy)$	0.993 $\pm$ 0.01	0.970 $\pm$ 0.03	0.949 $\pm$ 0.04	0.971 $\pm$ 0.05	0.934 $\pm$ 0.07	0.963 $\pm$ 0.06	0.992 $\pm$ 0.01	0.985 $\pm$ 0.03	0.957 $\pm$ 0.07	0.951 $\pm$ 0.05	0.967
UnNull	0.995 $\pm$ 0.01	0.963 $\pm$ 0.09	0.949 $\pm$ 0.09	0.976 $\pm$ 0.05	0.941 $\pm$ 0.08	0.968 $\pm$ 0.06	0.994 $\pm$ 0.06	0.986 $\pm$ 0.03	0.975 $\pm$ 0.05	0.958 $\pm$ 0.06	0.971
	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM	SAM
$Gy + f_\theta(Gy)$	0.185 $\pm$ 0.19	0.417 $\pm$ 0.31	0.434 $\pm$ 0.23	0.294 $\pm$ 0.19	0.428 $\pm$ 0.32	0.239 $\pm$ 0.16	0.076 $\pm$ 0.12	0.303 $\pm$ 0.20	0.353 $\pm$ 0.24	0.333 $\pm$ 0.27	0.306
$f_\theta(Gy)$	0.157 $\pm$ 0.15	0.325 $\pm$ 0.22	0.535 $\pm$ 0.23	0.278 $\pm$ 0.17	0.467 $\pm$ 0.34	0.189 $\pm$ 0.14	0.075 $\pm$ 0.11	0.276 $\pm$ 0.17	0.312 $\pm$ 0.18	0.285 $\pm$ 0.27	0.290
UnNull	0.158 $\pm$ 0.16	0.298 $\pm$ 0.20	0.413 $\pm$ 0.23	0.252 $\pm$ 0.17	0.398 $\pm$ 0.30	0.202 $\pm$ 0.13	0.062 $\pm$ 0.09	0.257 $\pm$ 0.16	0.334 $\pm$ 0.26	0.302 $\pm$ 0.25	0.268
	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS	ERGAS
$Gy + f_\theta(Gy)$	15.85	50.73	52.41	41.47	50.76	35.12	10.25	39.91	37.38	30.63	36.45
$f_\theta(Gy)$	13.98	46.53	54.13	40.10	51.09	26.03	09.51	34.99	39.25	23.87	33.95
UnNull	13.43	40.32	<u>45.23</u>	38.94	47.55	28.79	08.54	36.02	27.15	26.96	31.29

generalization ability of our method, we conducted additional experiments on various mosaic patterns, including traditional RGB Color Filter Arrays (CFAs) and different multispectral filter arrays (MSFAs). Specifically, we test: i) RGB Bayer CFA. The classical green-blue-red-green (GBRG) RGB Bayer filter array was used for demosaicing experiments; ii) IMEC multispectral filter arrays. Extra experiments were performed on 3×3 and 4×4 mosaic patterns, following the configurations described in [17]; iii) MSFAs proposed by Miao *et al.* [13] and Monno *et al.* [14]. These MSFAs leverage 4×4 multispectral filter arrays with the following configurations:

$$\begin{bmatrix} G & R & G & B \\ C & Y & C & Y \\ G & B & G & R \\ C & Y & C & Y \end{bmatrix}, \begin{bmatrix} G & R & G & Y \\ B & G & C & G \\ G & Y & G & R \\ C & G & B & G \end{bmatrix}. \quad (24)$$

Here, G, R, B, C, and Y represent green, red, blue, cyan, and yellow bands, respectively. Finally, five bands are obtained via these filter arrays.

The UnNull algorithm is designed as a generalized method independent of specific mosaic patterns. For the RGB Bayer CFA and the MFCAs proposed in [13] and [14], each pixel was treated as an individual channel during the initial reconstruction stage, producing 4 channels for the RGB CFA and 16 channels for the MFCAs. The final spectral bands were obtained by averaging the results of corresponding bands. For instance, the green band in RGB images was computed as the mean of the first and fourth demosaiced images. We have conducted experiments on the Scene 10 of CAVE dataset. The results of these experiments are summarized in Table VI. The UnNull algorithm demonstrated competitive performance across all tested mosaic patterns. These findings confirm the versatility of the proposed algorithm, making it effective not only for 25-band MFCAs but also for traditional RGB Bayer CFAs and more complex multispectral filter arrays.

Results on real-world datasets. In addition to experiments with synthetic data, we extend our method to two real-

world datasets captured using a real 5×5 and 4×4 IMEC camera [17]. The spatial size of the raw image is cropped to 1000×1000 for analysis and evaluation. Since ground truth data for the captured real scene is unavailable, we present only the raw images and reconstructed images generated by our UnNull and compared methods in Fig. 11. Specifically, we select the 10th spectral band for visualization. From the figure, it is evident that our UnNull algorithm excels in capturing finer and clearer image details and fewer mosaic patterns compared to other methods in both 5×5 and 4×4 IMEC camera cases. This observation underscores the effectiveness and robustness of our algorithm in real-world scenarios.

Results on spectral image reconstruction tasks. To further evaluate the performance and scalability of the proposed method, we conducted additional experiments on spectral image reconstruction tasks, specifically reconstructing hyperspectral images with 31 bands, where $B=31$ and $N_F=25$. We utilized the spectral response function of a 5×5 IMEC camera to simulate the matrix F in Eq. (5). The reconstruction results are summarized in Table VII. The proposed method demonstrates consistently high performance across all 31 bands. It achieves high PSNR and SSIM values while maintaining low SAM and ERGAS scores. These results indicate that the method has great potential for other spectral imaging systems.

Shortcomings of UnNull method and future works. While the UnNull method achieves significant reductions in computational time compared to previous approaches relying on untrained neural network priors, further optimization is needed, particularly for larger datasets and real-time applications. Incorporating appropriate regularization techniques about the scenes, such as information from previous frames, could further lower computational costs and enhance generalization capabilities. Moreover, although UnNull performs well in scenarios with moderate levels of missing data (*e.g.*, on a 5×5 mosaic camera), its accuracy tends to decline when handling irregular filter array patterns. Future research will focus on improving the method's robustness and accuracy in

these specific scenarios.

V. CONCLUSION

This paper proposes a novel and generalized Unsupervised range-Nullspace learning (UnNull) prior for spectral image reconstruction tasks. UnNull effectively decomposes the spectral image into its range and nullspace components, providing a more interpretable framework compared to existing supervised learning networks. Unlike previous unsupervised learning methods, our proposed UnNull algorithm not only achieves superior reconstruction performance but also offers high computational efficiency. Through comprehensive simulations and real-world experiments, we have demonstrated the performance, generalization ability, and efficiency of the UnNull method. Future research could explore further enhancements to UnNull and its applicability to other related tasks in image reconstruction.

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